Julia: A Language that Walks Like Python, Runs Like C

Julia is an emerging star in the world of programming languages. It offers the coveted combination of high performance and productivity. What Julia offers in terms of execution speed may almost seem unbelievable, especially to newcomers to this language. This article provides an introduction to Julia, which is as simple as Python to learn, and on par with C and Fortran in terms of execution speed.

As a programmer you may question the very need for yet another programming language. Such a question is very fair with the available ecosystem of a few hundred languages. However, each programming language is designed to achieve a specific goal. So it would be right to start this article by stating the specific objective of designing the Julia language. There is a basic conundrum in programming: execution speed vs development ease. If a language leans towards programmers, then there are likely to be issues with the execution speed. On the other hand, if you want to generate optimised code, then the development process will probably be harder. For example, a system completely designed in C programming language will generate optimal assembly code. However, development of such a system would take longer or the process might not be very friendly for a major section of programmers. Julia addresses these issues, as it combines the best of both worlds. It generates optimal machine code that runs on par with C programs, and at the same time the development process is equivalent to friendlier languages such as Python. The popular phrase in the Julia community is: "Walk Like Python; Run Like C."

Julia is an open source, high level programming language. It is a language for scientific, technical computing. You may be aware that scientific, technical computing problems are resource heavy. Julia is designed in a way that it allows developers to build solutions for scientific and technical computing problems swiftly, without compromising on performance. It comes loaded with an extensive library